Fewer molecules. Deeper experimental data.

AI may reduce candidate counts while increasing characterization per molecule

From molecule throughput to data density: a potential driver of life science tools demand

Existing data: staged screening

1,000 antibodies × 1 label

One experiment, one bit per molecule

Antibody	Binds?
Ab-001	Yes
Ab-002	No
Ab-003	Yes
Ab-004	Yes
Ab-005	No
Ab-006	Yes

What the model learns: will it bind

Everything else — stability, potency, manufacturability — is a guess

AI: richer, linked training data

1,000 antibodies × 9 label types

Same wet-lab campaign — nine measured properties per molecule

Antibody	KD	kon	koff	expr	Tm	agg	poly	epi	cell
Ab-001	0.82	0.64	0.71	0.55	0.48	0.39	0.30	0.58	0.44
Ab-002	0.35	0.28	0.41	0.22	0.61	0.52	0.44	—	0.18
Ab-003	0.68	0.75	0.59	0.47	0.33	0.28	0.62	0.35	0.51
Ab-004	0.91	0.58	0.83	0.66	0.72	0.58	0.24	0.49	0.76
Ab-005	0.44	0.31	0.38	0.58	0.41	0.66	0.71	—	0.29
Ab-006	0.57	0.69	0.52	0.73	0.55	0.44	0.37	0.62	0.58

low → high relative value per label (illustrative) — = not measured

What the model learns: nine properties and how they trade off — developability and potency, not binding alone

Reuse existing data

Normalize formats, conditions and access
Better training may need no new experiment

Stage 0: fill data gaps

Add missing assays and biological diversity
Create reusable model-training datasets

Active learning: measure selectively

Prioritize informative molecules and tests
Limit coverage of costly measurements

Terminology — what each of the nine labels measures

KD	overall binding strength; lower = tighter	kon	on-rate; how quickly binding forms	koff	off-rate; how long it stays bound
expression	how much antibody the cell makes	Tm	melting point; thermal stability	aggregation	self-clumping; risks immunogenicity
polyreactivity	sticks to off-targets; safety flag	epitope	where on the target it binds	cell potency	functional effect in living cells

Illustrative counts. Nine label types are not nine independent assays; richer data need not come at unchanged cost.

Sources: Jain et al., 2017; Arsiwala et al., 2025; van Niekerk et al., 2026. Mechanism illustration; definitions and links in SOURCES.md.